

Electronic structure and magnetic and electromagnetic wave absorption properties of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ M-type hexaferrites

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ABSTRACT

Crystal, electronic structures and the magnetic and electromagnetic wave absorption properties of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ ($x = 0-2$) M-type hexaferrites prepared by a co-precipitation technique were studied. The analyses of X-ray diffraction patterns indicated that the samples mainly crystallized in the $P6_3/mmc$ hexagonal structure, with the additional constitution of Y-type hexaferrite as $x > 0$. The replacement of Co^{2+} for Fe^{3+} in $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ changed the lattice constants and caused lattice distortions. Particularly, Co^{2+} doping also reduced magnetization and hard magnetic property of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$. This is ascribed to magnetic moment of Co^{2+} smaller than that of Fe^{3+} and to the decrease of magnetocrystalline anisotropy. Having studied electromagnetic wave absorption properties in the frequency range $f = 0.1-18$ GHz, we found $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ showing high reflection loss (RL) values at frequencies of 0.1–15 GHz, but fairly low RL values at higher frequencies. These features suggest that $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ can be suitable to electronic devices working at GHz frequencies.

1. Introduction

Hexaferrites are iron oxides with a hexagonal structure. Apart from Fe and O ions, these compounds have one or more other elements that could be Ba, Sr, and/or Co. Basing on the Fe/O ratio and chemical elements, it has been classified hexaferrites into six groups, namely M-, Z-, Y-, W-, X-, and U-type hexaferrites [1,2]. With tunable magnetic properties, high Curie temperature, high corrosion resistance, and low-cost production, hexaferrites have been widely used in many appliances, such as loud speakers, electricity-generating wind turbines, electromechanical machines, and electrical transformers. Recently, hexaferrites have also been proven to be potential materials for electronic devices working in the GHz frequency range [3–5].

Among hexaferrites, M-type hexaferrite has the simplest structure of $\text{MeFe}_{12}\text{O}_{19}$, with $\text{Me} = \text{Ba}, \text{Sr}, \text{Pb}, \text{etc.}$ Its unit cell is composed of the form SRS^*R^* , in which S is the spinel block and R is the hexagonal block. The blocks S^* and R^* have the atomic arrangements same as S and R, respectively, but rotated at 180° along the c -axis. Ferromagnetic (FM) order is generated from the ferrimagnetically collinear structure of 24 Fe^{3+} ions occupying three FeO_6 octahedral 2a, 4f₂ and 12k sites, one FeO_4 tetrahedral 4f₁ site, and one FeO_5 trigonal-bipyramidal 2b site. It should be noticed that 16 Fe^{3+} ions with the spin up direction at the 2a, 2b and 12k sites and eight Fe^{3+} ions with the spin down direction at the 4f₁ and 4f₂ sites would produce a large net magnetic

moment and large magnetocrystalline anisotropy. Due to the occupancy of Fe^{3+} ions in FeO_4 , FeO_5 , and FeO_6 polyhedra, there are many possible exchange interactions [6]. Their interaction strength between Fe^{3+} ions depend on Fe–O distances and Fe–O–Fe bond angles. The modification of these bond distances and angles (upon replacing Me^{2+} and/or Fe^{3+} ions by other elements) is necessary to control the magnetic and magnetoelectric properties of M-type hexaferrites for application demands. Basically, one can replace Me^{2+} by an alkaline-earth/rare-earth metal, Fe^{3+} by a transition metal, or the combination of these.

For $\text{BaFe}_{12}\text{O}_{19}$ M-type hexaferrite, the doping of a divalent alkaline-earth ion, the change in the magnetoelectric properties is mainly related to lattice-strain effects because the valency of Fe ($3+$) is unchanged [7,8]. However, the substitution of a RE^{3+} ion ($\text{RE} = \text{rare-earth element}$) for Ba^{2+} leads to a coexistence of Fe^{2+} and Fe^{3+} because of the charge balance [9]. Accordingly, the magnetoelectric properties would be changed by not only the lattice-strain effects, but also the magnetic interactions between $\text{Fe}^{2+,3+}$ ions occupying at the different sites. For the case of $\text{BaFe}_{12}\text{O}_{19}$ doped with a transition metal (TM), the valency and position of TM as well as magnetic interactions between TM and Fe ions are still debate topics. It has been suggested that Fe^{3+} is stable while the valency of TM is $2+$ as $\text{TM} = \text{Co}, \text{Zn}, \text{Ni}, \text{or Cu}$ [10–16], and $4+$ as $\text{TM} = \text{Ti}, \text{Zr}, \text{or Ir}$ [11,14,17]. Though many research works have carried out on TM-doped $\text{BaFe}_{12}\text{O}_{19}$ materials, the

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problems relative to the oxidation number and the occupancy of TM ions in the $\text{BaFe}_{12}\text{O}_{19}$ host lattice and the electromagnetic wave absorption properties at GHz frequencies have not been fully understood.

In this work, we have prepared $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ compounds and studied the crystalline and electronic structures and the magnetic and electromagnetic wave absorption properties, using experimental tools of a scanning electron microscope, X-ray diffractometer, X-ray absorption spectroscopy, a vibrating sample magnetometer, and a network analyzer. Our study points out that $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ compounds have the coexistence of Fe^{3+} and Co^{2+} ions. Though the Co^{2+} doping reduced the hard magnetic property of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$, the electromagnetic wave absorption (EMA) properties in the frequency range $f = 0.1\text{--}15\text{ GHz}$ were less changed, with reflection loss (RL) values of $|RL| < 5\text{ dB}$. At higher frequencies, however, $|RL|$ rapidly increased as $x > 0$. These features suggest potential applications of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ materials in electronic devices working at GHz frequencies.

2. Experimental details

$\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples with $x = 0, 0.5, 1, 1.5$ and 2 were prepared by a simple co-precipitation method. Sigma-Aldrich chemicals (including $\text{Ba}(\text{NO}_3)_2$, $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, and $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, purity of 99%) with nominal compositions of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ were dissolved in 100 mL deionized (DI) water at 90°C . After dissolving, 50 mL NaOH solution (1M) was added to the mixtures and continuously stirred at 90°C for 2 h. When chemical reactions were finished, precipitation was filtered, washed by ethanol, and dried at 100°C . Dried powders were then annealed at 900°C for 3 h in air. All final products obtained after annealing were checked the morphology by a scanning electron microscope (SEM, JSM-5410LV). The crystal structure was examined by a Rigaku MiniFlex X-ray diffractometer working with a $\text{Cu-K}\alpha$ radiation source ($\lambda = 1.5406\text{ \AA}$). The electronic and local geometrical structures were studied by using X-ray absorption spectroscopy (XAS) at the 8C Nano-XAFS Beamline of the Pohang accelerator laboratory, South Korea. XAS spectra in the transmission configuration were measured for the Fe *K*-edge ($E_0 = 7112\text{ eV}$) and Co *K*-edge ($E_0 = 7709\text{ eV}$). Some Fe and Co oxides were also used as reference. The magnetic properties at temperatures $T = 15\text{--}300\text{ K}$ were studied by using a vibrating sample magnetometer (VSM 7404, Lake Shore Cryotronics, Inc), where the magnetic field (H) magnitude could be changed within 10 kOe. To investigate the EMA properties (through measurements of the complex permeability, μ_r , and permittivity, ϵ_r), we shaped typical hexaferrite powders ($x = 0, 1$, and 2) into toroids as follows. Selected hexaferrites combined with 7 wt% epoxy were well ground and pressed into a toroid (the outer and inner diameters were 7 and 3 mm, respectively, and the thickness (d) was fixed at 4 mm). After that, all toroids dried at 170°C for 30 min were studied the EMA properties at room temperature (RT), using a PNA-X Network analyzer working at frequencies $f = 0.1\text{--}18\text{ GHz}$.

3. Results and discussion

Following fabrication, we checked the particle morphology of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples using an electron microscope. The micrographs of typical $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples ($x = 0, 0.5, 1$, and 2) shown in Fig. 1(a–d) exhibit that the samples consist of platelet- and flake-shaped particles with sizes of 100–200 nm. Their crystal orientation in the samples is random. There are also interstices between particles, and the number of interstices tends to decrease with increasing x .

To check the crystal structure, we recorded X-ray diffraction (XRD) patterns of all fabricated $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples in the angle range $2\theta = 20\text{--}70^\circ$, with scanning steps fixed at 0.01° , as shown in Fig. 2. Careful analyses of the XRD patterns indicated that pure $\text{BaFe}_{12}\text{O}_{19}$ ($x = 0$) fully crystallized in the $P6_3/mmc$ M-type hexagonal structure, in accordance with a standard card JCPDS#043-0002. For $x > 0$, however, besides the main phase of M-type hexaferrite, there is the addition

of a secondary phase of Y-type hexaferrite ($\text{Ba}_2\text{Co}_2\text{Fe}_{12}\text{O}_{22}$). This phase gradually develops when x increases from 0.5 to 2, as asterisked in Fig. 1. In fact, the formation of secondary phases as synthesizing a specific hexaferrites usually takes place because these secondary phases can also constitute under the same fabrication conditions [1,18,19]. From the XRD data, we could calculate the average crystallite size (d) upon the Scherrer equation of $\beta \cos \theta = K\lambda/d$, in which $K = 0.89$ is a factor associated with the crystallite shape and β is the width of a diffraction angle 2θ . As shown in Table 1, d gradually decreases from 37.9 nm for $x = 0$ to about 26.7 nm for $x = 1.5$, but again rapidly increases to 38.2 nm for $x = 2$.

Together with identifying the structural phases in $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$, we also calculated the lattice parameters (a and c) of the $P6_3/mmc$ M-type hexagonal structure. The data shown in Fig. 3 reveal that while a tends to decrease, c gradually increases with increasing x . This result demonstrates the elongation of the c -axis and the shrinking of the a -axis. To figure out the nature of the lattice-distortion phenomenon, we have taken into account for the electronic structure of Fe and Co in $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$. Fig. 4(a) shows Fe *K*-edge XAS spectra of typical samples with $x = 0, 1$, and 2 comparing with the spectrum of Fe_2O_3 . It comes to our attention that the absorption edges of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ and Fe_2O_3 overlap each other at $\sim 7123\text{ eV}$. Additionally, if considering the Co *K*-edge XAS spectra, as shown in Fig. 4(b), one can see that the absorption edges of the typical samples with $x = 1$ and 2 coincide with the absorption edge of CoO at $\sim 7717\text{ eV}$. These results reflect that only Fe^{3+} ($3d^5$) and Co^{2+} ($3d^7$) ions exist in our $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples, in good with those reported previously on Co-doped $\text{BaFe}_{12}\text{O}_{19}$ hexaferrites [1,15], in which research groups used other techniques.

We could also estimate the average bond distances from Fe to O in $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ through the Fourier transform (FT) of the Fe *K*-edge XAS spectra, meaning the consideration of the radial distribution function (R) around the central absorbing Fe atom. The FT spectra of the typical samples are graphed in Fig. 5. The strongest peak at $\sim 1.4\text{ \AA}$ corresponds to an average distance from Fe to the nearest neighbours (labelled as $R_{\text{Fe-O}}$), which are oxygen ions in the FeO_4 , FeO_5 , and FeO_6 polyhedra. In fact, owing to the phase shift of photoelectron back-scattering, R values in Fig. 7 were shifted by $\sim 0.5\text{ \AA}$ from their true values [20]. As shown in Table 1, the values of $R_{\text{Fe-O}}$ are about 1.93, 1.89, and $1.87\text{ \AA}$ for $x = 0, 1$, and 2 , respectively. A gradual decrease of $R_{\text{Fe-O}}$ with increasing x is similar to the variation tendency of a . We think that the Co doping caused the shrinking of the a -axis and the elongation of the c -axis, leading to the opposite variation of a and c , as plotted in Fig. 3. Under the lattice-distortion impact, more O atoms close to the absorbing Fe atom would contribute to extended X-ray absorption fine structure (EXAFS) signals (which are the sum of all outgoing and incoming waves and dependent on the environment surrounding the absorbing atom [21]), and cause the shift of the $R_{\text{Fe-O}}$ peak towards shorter distances when x increases. The lattice distortion is due to the substitution of Co^{2+} with a larger ionic radius (r_{ion}) for Fe^{3+} with a smaller r_{ion} value. Though Co^{2+} ions prefers tetrahedral ($4f_1$) sites, they are also ascribed to occupy randomly the trigonal-bipyramidal ($2b$) and octahedral ($2a$, $4f_2$, $12k$) sites. Depending on the spin state and sites of Co^{2+} and Fe^{3+} ions in the polyhedra, the r_{ion} values would be different. For the high spin state and the coordination numbers $N = 4\text{--}6$, r_{ion} values are about $0.49\text{--}0.645\text{ \AA}$ for Fe^{3+} and $0.58\text{--}0.74\text{ \AA}$ for Co^{2+} [22].

To study the influence of the Co^{2+} doping on the magnetic properties of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$, we recorded $M(H)$ hysteresis loops at some temperature points in the range $T = 15\text{--}300\text{ K}$. Fig. 6(a and b) plot representatively the $M(H)$ data at 15 and 300 K. It comes to our attention that all the samples show large hysteresis loops characteristic of hard magnetic materials. Different from conventional ferromagnets with smooth hysteresis loops [23], the loop shape of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ is anomalous at low fields. This phenomenon is thought to be the presence of Y-type hexaferrite (as identified by XRD), which has the coercivity (H_c) much smaller than that of M-type hexaferrite [1,24]. Although H

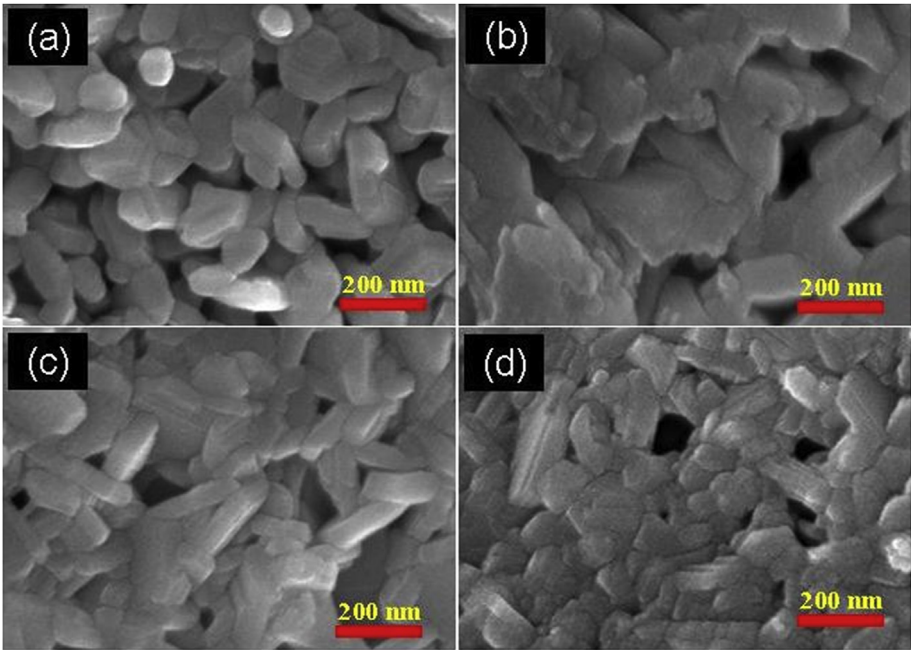


Fig. 1. Micrographs of BaFe_{12-x}Co_xO₁₉ samples with (a) $x = 0$, (b) $x = 0.5$, (c) $x = 1$, and (d) $x = 2$.

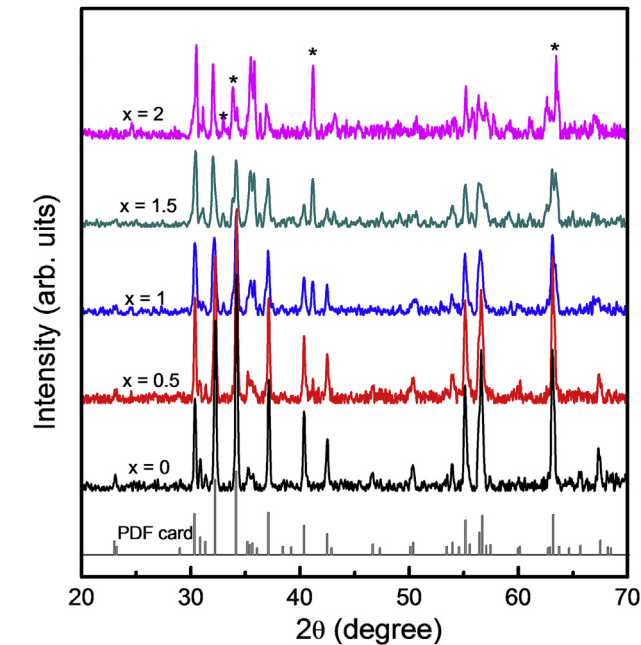


Fig. 2. XRD patterns of all BaFe_{12-x}Co_xO₁₉ samples ($x = 0-2$) comparing to a standard pattern (PDF card, JCPDS#043-0002). Asterisks labelled for $x > 0$ indicate the diffraction peaks coming from a secondary phase of Y-type hexaferrite.

Table 1

Experimental values obtained from analyzing XRD patterns, EXAFS spectra, and magnetic hysteresis loops (at 300 K) of BaFe_{12-x}Co_xO₁₉ samples.

x	d (nm)	$R_{\text{Fe-O}}$ (Å)	$b/10^6$ (Oe ²)	H_a (kOe)	$K_1/10^6$ (erg/cm ³)
0	37.9	1.93	10.68	12.66	2.08
0.5	37.6	—	8.28	11.14	1.68
1	27.4	1.89	6.12	9.59	1.29
1.5	26.7	—	4.93	8.60	1.14
2	38.2	1.87	4.29	8.02	0.97

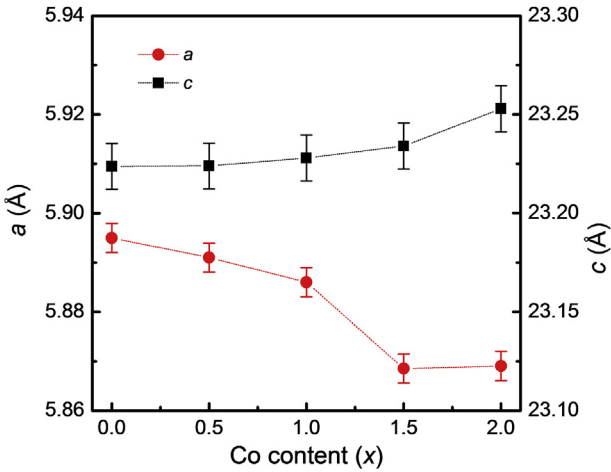


Fig. 3. Variations of the lattice constants (a) a and (b) c as a function of x in BaFe_{12-x}Co_xO₁₉.

increases up to 10 kOe, M does not achieve the saturation magnetization (M_s). This feature is popular for hexaferrites because they exhibit short-range ferromagnetic (FM) order with complex exchange interactions between TM ions at different polyhedral sites. From the recorded $M(H)$ data, we determined M_s and H_c values versus temperature T of all samples, as graphed in Fig. 7(a and b). For M_s at a given T , its value decreases with increasing x , and a rapid reduction of M_s is observed for $x = 2$. With increasing T from 15 to 300 K, M_s of the samples $x = 0-1.5$ gradually decreases. Meanwhile, M_s of $x = 2$ firstly increases as T changes from 15 to 100 K, keeps fairly stable as $T = 100-250$ K, and then slightly decreases as $T > 250$ K. Such anomalous $M_s(T)$ variation of $x = 2$ is ascribed to the presence of Y-type hexaferrite that could cause the magnetic-interaction competition with the M-type hexaferrite phase. For all samples, the decrease of M_s with increasing T is due to the decline of FM order under the impact of thermal energy, like the cases of BaFe_{12-x}Mn_xO₁₉ [25] and BaFe_{12-x}Al_xO₁₉ [26]. Meanwhile, the decrease of M_s with increasing x in BaFe_{12-x}Co_xO₁₉ is due to the magnetic moment of Co²⁺ smaller than that of Fe³⁺ [13]. Additionally, the Co²⁺ substitution caused the lattice distortion and spin canting, and

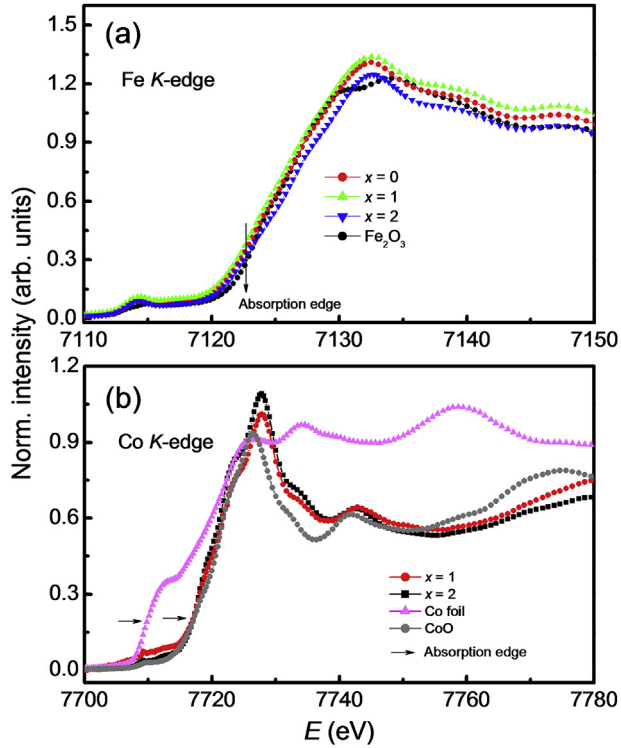


Fig. 4. XAS spectra recorded at (a) Fe K- and (b) Co K-edge for typical $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples.

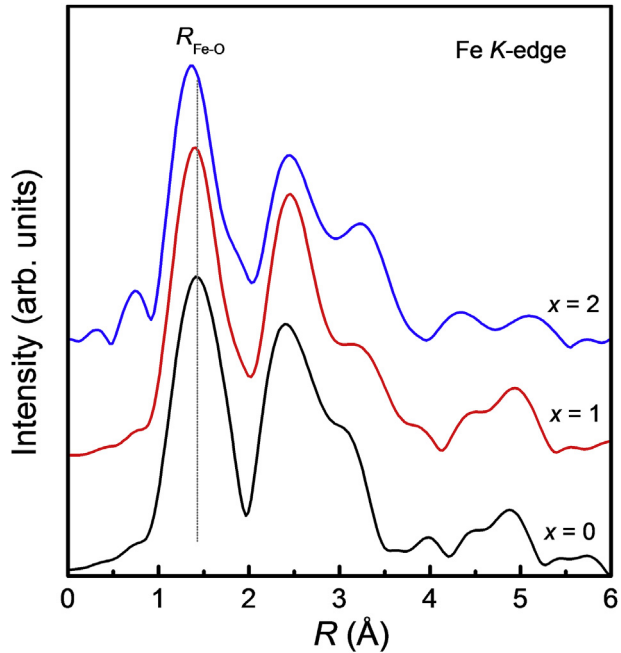


Fig. 5. Fe K-edge FT XAS spectra of typical $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples with $x = 0, 1$, and 2 .

weakened exchange interactions of $\text{Fe}^{3+}\text{-O-Fe}^{3+}$ pairs [27]. These factors broke the collinear magnetic structure of Fe^{3+} ions and thus reduced M_s .

Similar to M_s , H_c also varies as function of T , see Fig. 7(b). Comparing with $x = 0$, the H_c values of the samples $x = 0.5\text{--}2$ are much lower. Their H_c values are less changed at temperatures $T < 100$ K, but become decreased with increasing T and x . For each sample, its T -dependent H_c is relative to the dynamics of magnetocrystalline anisotropy,

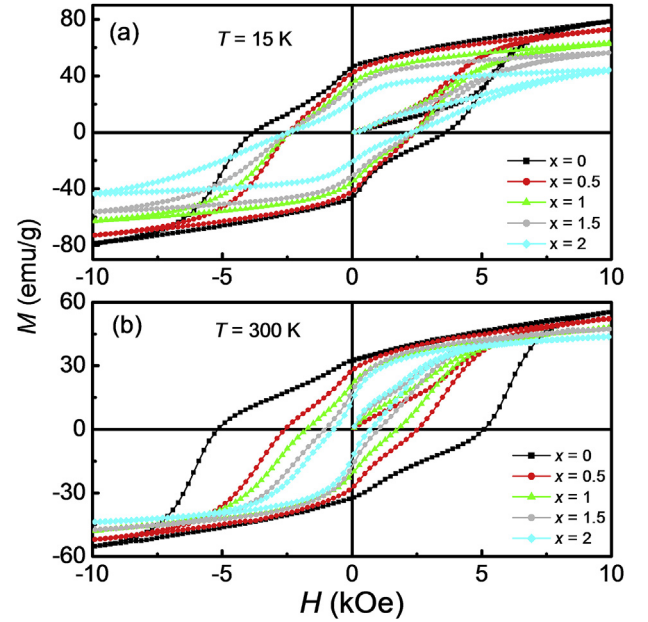


Fig. 6. $M(H)$ hysteresis loops of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples at (a) 15 K and (b) 300 K.

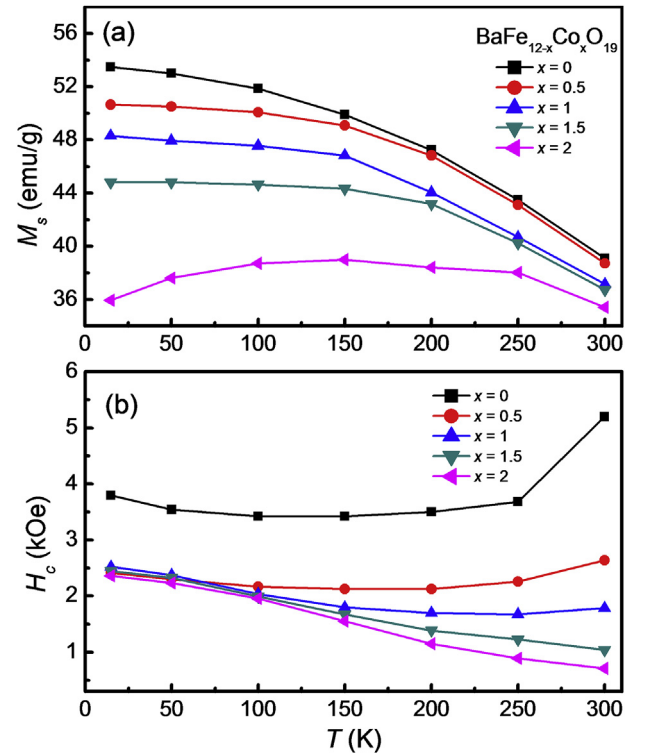


Fig. 7. Variation of (a) M_s and (b) H_c as a function of x and temperature.

crystal lattice, and magnetic domain, which are also varied according temperature. Meanwhile, the H_c change for different samples at a specific T value is mainly related to magnetocrystalline anisotropy characterized by the anisotropy field (H_a , which is the sum of magnetocrystalline anisotropy, shape anisotropy and stress anisotropy) and anisotropy constant (K_1). These parameters can be calculated from M_s upon the following relations [8].

$$M = M_s \left(1 - \frac{b}{H^2} \right), \quad (1)$$

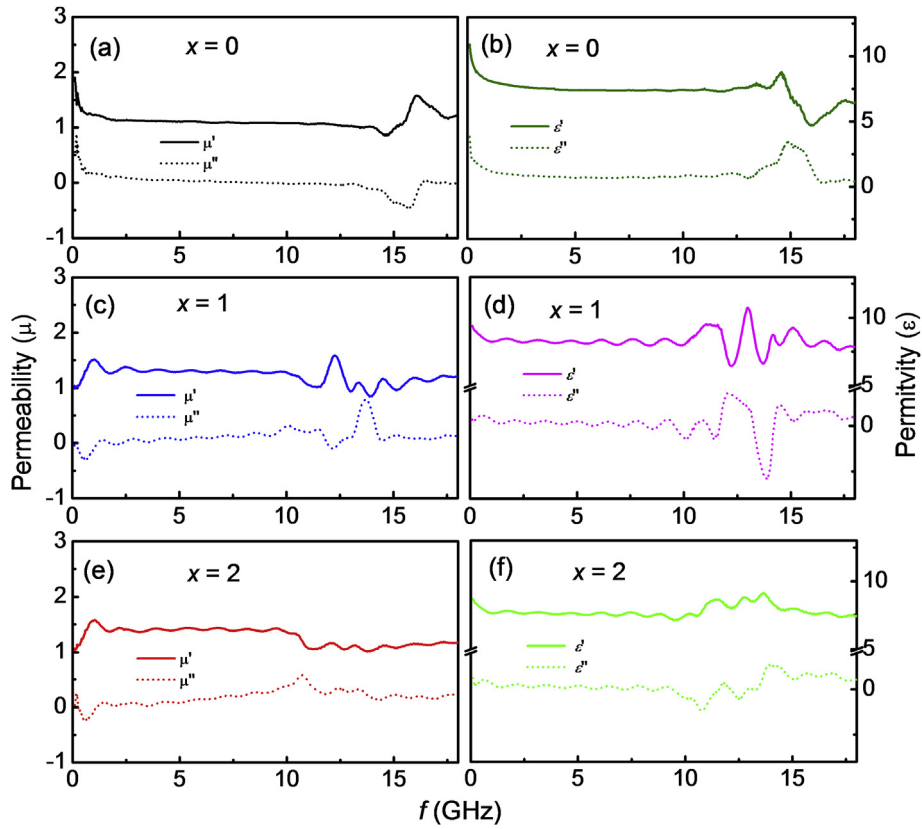


Fig. 8. Frequency dependences of the complex permeability and permittivity of typical $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples with (a) $x = 0$, (b) $x = 1$, and (c) $x = 2$ obtained for thickness $d = 4$ mm.

$$b = \frac{H_a^2}{15} = \frac{4K_1^2}{15M_s^2}, \quad (2)$$

in which b is a parameter dependent on magnetocrystalline anisotropy. In Table 1, it shows representatively the values of b , H_a , and K_1 obtained at 300 K. One can see that both parameters H_a and K_1 decrease with increasing x . It means that the reduced magnetocrystalline anisotropy in $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ as increasing x caused the reduction of H_c .

Following the structural, electronic and magnetic studies, the EMA properties of three typical $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples ($x = 0, 1$, and 2) at RT in the frequency range $f = 0.1$ – 18 GHz were investigated. Because the EMA is related on the complex permeability ($\mu_r = \mu' - i\mu''$) and permittivity ($\epsilon_r = \epsilon' - i\epsilon''$), where μ' and ϵ' are the real part (associated with the storage capability of electric and magnetic energy) while μ'' and ϵ'' are the imaginary one (associated with the loss of electric and magnetic energy) [28], we measured these f -dependent parameters. The results obtained are shown in Fig. 8(a–f). In regard to the permeability, μ' of $x = 0$ in the range $f = 1$ – 12 GHz is less changed ($\mu' \approx 1.06$), but becomes large at frequencies of ~ 0.21 GHz ($\mu' = 1.8$) and 16.5 GHz ($\mu' = 1.6$). The enhancement of μ' at these frequencies is attributed domain-wall and FM resonances. These processes cause magnetic losses, leading to the fact that the variation of μ'' is consistent with that of μ' at corresponding frequencies, see Fig. 8(a). For the Co-doped samples with $x = 1$ and 2 , their μ' values (1.3 – 1.4) are larger than those of $x = 0$, particularly at frequencies $f < 11$ GHz, see Fig. 8(c, e). Around 11 – 15 GHz, there are few peaks with different magnitudes observed for both μ' and μ'' . These peaks are assigned to FM resonance signals related to the additional presence of Co^{2+} that causes the exchange interactions between Fe^{3+} and Co^{2+} ions [29] and the fluctuation of spins in $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ hexaferrites. In other words, the FM resonance peaks of the Co-doped samples were shifted towards lower frequencies if comparing with $x = 0$ (in which there are only exchange interactions between Fe^{3+} ions). The shift of the resonance frequency

(f_r) is assigned to the reduction of H_a because a relation between f_r and H_a is given by $f_r = (\gamma/2\pi)H_a$, where γ is the gyromagnetic ratio [17].

If considering the permittivity, we have also observed the variation tendencies of ϵ' and ϵ'' fairly similar to those of μ' and μ'' at corresponding frequencies, see Fig. 8(b, d, f). The resonance is observed at ~ 15 GHz for $x = 0$, but at lower frequencies (< 15 GHz) for $x = 1$ and 2 . Interestingly, ϵ' values of $x = 1$ and 2 at $f > 0.5$ GHz are greater than those of $x = 0$. In the range $f = 10$ – 15 GHz, some ϵ'' values of $x = 1$ and 2 are negative while those of $x = 0$ is always positive, which is different from previous reports on $\text{BaFe}_{12}\text{O}_{19}$ and $\text{SrFe}_{12}\text{O}_{19}$ based M-type hexaferrites [17,27,30,31]. The negative ϵ'' phenomenon is usually observed in metamaterials [32], in which one been suggested that it is correlated to the magnetoelectric coupling of multiferroic materials [33].

All of such variations of the permeability and permittivity influence the EMA properties of $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$. Theoretically, the EMA properties are characterized by the parameter RL that is calculated upon the following equations [17,27,31].

$$Z_{in} = Z_0 \sqrt{\frac{\mu_r}{\epsilon_r}} \tanh \left[i \frac{2\pi d f}{c} \sqrt{\mu_r \epsilon_r} \right], \quad (3)$$

$$RL = 20 \lg \left| \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \right|, \quad (4)$$

where Z_{in} is the input impedance and Z_0 ($\approx 377 \Omega$) is the characteristic impedance of vacuum. With $d = 4$ mm and $f = 0.1$ – 18 GHz, we calculated $RL(f)$ values of the typical $\text{BaFe}_{12-x}\text{Co}_x\text{O}_{19}$ samples ($x = 0, 1$, and 2). As shown in Fig. 9, all samples show RL values larger than -5 dB at frequencies $f = 0.1$ – 15 GHz. However, for higher frequencies ($f = 15$ – 18 GHz), while RL of $x = 0$ remains high, that of $x = 1$ and 2 rapidly decreases, about -22.3 and -12.7 dB at $f = 18$ GHz for $x = 1$ and 2 , respectively. Due to the limit of experimental equipment, we

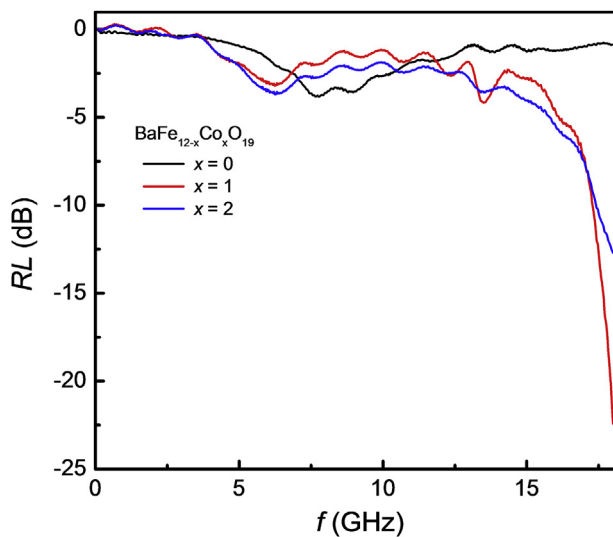


Fig. 9. Frequency-dependent RL data of typical $BaFe_{12-x}Co_xO_{19}$ samples with $x = 0, 1$, and 2 .

could not carry out measurements at higher frequencies to find minimum RL values of these samples. It should be noticed that the value $RL = -22.3$ dB of $x = 1$ achieved at 18 GHz is fairly small, corresponding to $\sim 99\%$ of incident microwaves being absorbed. A larger RL value (-12.7 dB) obtained for $x = 2$ at 18 GHz could be due to a rapid decrease of M_s and/or the additional presence of Y-type hexaferrite. The EMA nature is due to the fact that electromagnetic wave energy can be absorbed and dissipated into heat through dielectric and magnetic losses [31]. For the magnetic loss, it is associated with domain-wall and FM resonances induced by the interaction between electromagnetic waves and magnetic moments of Fe^{3+} and Co^{2+} ions in $BaFe_{12-x}Co_xO_{19}$ hexaferrites. Meanwhile, the dielectric loss versus frequency is mainly associated with electron hopping between Fe^{3+} and Co^{2+} ions (i.e., the electric dipole polarization) and the surface state and grain surface energy (the interfacial polarization) [27]. Clearly, while pure $BaFe_{12}O_{19}$ ($x = 0$) is not useful for the EMA, $BaFe_{12-x}Co_xO_{19}$ compounds with a suitable x value could improve remarkably the EMA properties at high frequencies $f > 15$ GHz. Studying Zr-doped $BaFe_{12}O_{19}$ hexaferrites [17,27], one also found the enhanced EMA possibility at $f > 15$ GHz, with $|RL| = 20\text{--}30$ dB. The co-doping of more elements with different valences (such as $BaCo_xCu_xZr_{2-x}Fe_{12-4x}O_{19}$) [31] or the fabrication of $CaCoTi$ ferrite composites [34] could improve the EMA properties at lower frequencies, in the range $f = 9\text{--}15$ GHz. It means that it is possible to tune the EMA range of M-type hexaferrite as expected upon the co-doping and compositing.

4. Conclusion

In this work, we prepared polycrystalline M-type hexaferrites of $BaFe_{12-x}Co_xO_{19}$ ($x = 0\text{--}2$). The XRD analyses indicated the samples mainly crystallizing in the $P6_3/mmc$ structure, with the addition of a secondary phase as $x > 0$. The XAS study proved that the oxidation numbers of Fe and Co are $2+$ and $3+$, respectively, which are unchanged versus x . Particularly, the replacement of Co^{2+} for Fe^{3+} in $BaFe_{12-x}Co_xO_{19}$ caused the elongation of the c -axis and the shrinking of the a -axis. Concerning the magnetic properties at room temperature, the Co^{2+} doping reduced M_s . This is ascribed to the magnetic moment of Co^{2+} smaller than that of Fe^{3+} . We believe that the Co^{2+} substitution with a smaller magnetic moment (comparing with that of Fe^{3+}) caused the lattice distortion and spin canting, and weakened exchange

interactions of $Fe^{3+}\text{--}O\text{--}Fe^{3+}$ pairs. These factors broke the collinear magnetic structure of Fe^{3+} ions and reduced M_s . Concurrently, and H_c also decreases with increasing x because of the reduction of magneto-crystalline anisotropy. Investing the EMA properties, we found $BaFe_{12-x}Co_xO_{19}$ samples with $x = 1$ and 2 exhibiting large RL losses (> -5 dB) at frequencies $f = 0.1\text{--}15$ GHz, but low RL values at high frequencies (< -12 dB). Meanwhile, pure $BaFe_{12}O_{19}$ ($x = 0$) does not exhibit the EMA in the studying f range. These results reflect that the selection of $BaFe_{12-x}Co_xO_{19}$ with a suitable x value could improve the EMA properties at high frequencies $f > 15$ GHz.

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